



Answer the following questions:

(Total marks: 75)

1st Question

Marks: 30

[2 Marks each] State whether each of the following statements is True (T) or False (F)

1. Image pre-processing increases the image information content ()
2. Pixel brightness transformation is one of image pre-processing tasks ()
3. Histogram equalization changes the intensity levels of an input image to spread the output image histogram equally across all intensity levels ()
4. Number of pixels in an output image after applying pixel co-ordinate transformation is equal to the number of pixels in input image if the value of Jacobian $J > 1$ ()
5. Convex hull has at least two points that are connected by line that is completely inside the convex hull ()
6. CMY color system is a subtractive color mixing system ()
7. Original image can be reconstructed from the integral image ()
8. Image pre-processing may change intensity levels of the image pixels ()
9. Image convolution may replicate image borders to calculate the edge pixels ()
10. Median filter radius must be even number ()
11. The direction of the edge at location (x, y) is perpendicular to the gradient vector at that point ()
12. Harris cornet detector is invariant to large scale changes ()
13. Adaptive thresholding is a segmentation technique that divides the image into equal regions and apply global thresholding to each sub region separately ()
14. Semi-thresholding masks out the background and keeps the other segments of the image without changing their intensity levels. ()
15. Global thresholding optimum threshold can be easily detected using image histogram ()

2nd Question

Marks: 15

[3 Marks each] Select the correct answer

1. D8 Distance of two pixels on the same diagonal is _____.
 - a. Number of rows
 - b. Number of columns
 - c. A or B
 - d. None of Previous
2. Example of non-linear color space is _____.
 - a. RGB
 - b. CMY
 - c. HSV
 - d. A or B

3. Applying Laplacian filter to an image to highlights _____.

- a. Edges
- b. Noise
- c. Continuity
- d. All the previous

4. Applying Gaussian kernel to Laplacian filter will result in _____.

- a. Smoothed edges
- b. Sharp Image
- c. Blurry Image
- d. None of Previous

5. Averaging filter which has equal weights can be calculated efficiently by using _____.

- a. Histogram
- b. Integral Image
- c. Co-occurrence Matrix
- d. None of Previous

3rd Question

Marks: 15

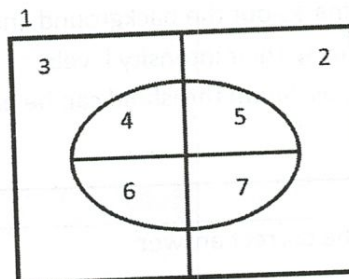
(5 Marks)

A. Write the run length code for the binary image below:

	0	1	2	3	4	5
0						
1						
2						
3						
4						
5						

B. Draw Regional Adjacency graph for the image below:

(5 Marks)



C. Calculate the integral image for the image below:

(5 Marks)

6	1	0	4
0	6	5	9
0	5	1	4
5	9	4	6



4th Question

Marks: 15

A. Apply Laplacian transform to the image below:

(10 Marks)

Note that the operator for the Laplacian is

0	-1	0
-1	5	-1
0	-1	0

6	4	9	5
4	1	5	0
9	5	6	0
4	0	1	6

B. Apply Global Thresholding to the image below:

(5 Marks)

Note that the threshold value is 10.

6	10	20	5
1	6	5	19
0	15	1	14
4	9	4	15

Good luck

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